Edexcel A-Level Maths · Paper 1: Pure Mathematics 1 · 9MA0/01

Pure 1 — Full Worked Solutions

Wednesday 3 June 2026 · Advanced GCE Level 3

Sat the paper and want to know how you did? Here are **StudyHours model solutions** to all 15 questions — every method step shown the way an examiner wants it, so you can mark your own script and see exactly where the marks were. Built by tutors, not a calculator.

All 15 questions, full method

Answers boxed in green. Common slips flagged in red. “Show all working” questions worked without calculator shortcuts.

1 Transformations of $y = f(x)$, point P(−4, 5)

4 marks

(a) y = 4f(2x)

$2x$ inside → horizontal stretch scale factor ½: x-coord halved. $\times 4$ outside → vertical stretch scale factor 4: y-coord ×4.

$x: -4 \to -2 \quad y: 5 \to 20$

(−2, 20)

(b) y = f(x + 3) − 2

$x+3$ → translate left 3: $-4 \to -7$. -2 → translate down 2: $5 \to 3$.

(−7, 3)

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2 Iteration · $f(x) = e^{2x-1} + 3x - 7$

5 marks

(a) Show α lies in [1.1, 1.2]

$f(1.1) = e^{1.2} + 3.3 - 7 = 3.320 - 3.7 = -0.380 < 0$

$f(1.2) = e^{1.4} + 3.6 - 7 = 4.055 - 3.4 = +0.655 > 0$

Sign change and f is continuous → a root α lies in [1.1, 1.2]. ∎

(b) $x_{n+1} = ½(1 + \ln(7 - 3x_n))$, $x_1 = 1.1$

$x_2 = ½(1 + \ln(7 - 3.3)) = ½(1 + \ln 3.7)$

$x_2 = 1.1542$ (4 d.p.)

Continue iterating ($x_3 = 1.1318$, $x_4 = 1.1411$, … converging):

α = 1.1384 (4 d.p.)

3 Differentiation & tangent · $y = x^3 - 2x^2 + 5x - 7$ 5 marks

(a) Find dy/dx

$dy/dx = 3x^2 - 4x + 5$

(b) Tangent at x = 2

y-coord: $8 - 8 + 10 - 7 = 3$ → point (2, 3).

Gradient: $3(4) - 4(2) + 5 = 12 - 8 + 5 = 9$.

$y - 3 = 9(x - 2)$

$y = 9x - 15$ (m = 9, c = −15)

4 Functions · $f(x) = 3x^2 - 2$, $g(x) = (5x - 1)/(x - 4)$ 5 marks

(a) Range of f

Min at x = 0 gives −2; parabola opens up. $f(x) \geq -2$

(b) gf(2)

$f(2) = 12 - 2 = 10$, then $g(10) = (50 - 1)/(10 - 4) = 49/6$

$gf(2) = 49/6$

(c) $g^{-1}(x)$

Let $y = (5x-1)/(x-4)$ → $y(x-4) = 5x-1$ → $yx - 5x = 4y - 1$ → $x = (4y-1)/(y-5)$

$g^{-1}(x) = (4x - 1)/(x - 5)$

5 Parametric model · $x = 100\sqrt{t}$, $y = 10 \sin(t^2/4)$ 5 marks

(a) Maximum height

Max of $10 \sin(\ldots)$ is when sin = 1. 10 m

(b) Horizontal distance when y = 5 the second time

$10 \sin(t^2/4) = 5 \rightarrow \sin(t^2/4) = ½$

Solutions of $t^2/4$: $\pi/6$ (1st time), $5\pi/6$ (2nd time).

$t^2 = 4\cdot(5\pi/6) = 10\pi/3 \rightarrow t = \sqrt{(10\pi/3)}$

$x = 100\sqrt{t} = 100\cdot(10\pi/3)^{1/4}$

$x \approx 179.9$ m

SLIP Second time is on the way down: $t^2/4 = 5\pi/6$, not the next full cycle. Easy to take the wrong root.

6 Integration · $\int 3/(7x + 1)\,dx$ — 5 marks

(a)

$(3/7)\ln|7x + 1| + C$

(b) Find p where $\int_2^p = 6/7$

$(3/7)[\ln(7p+1) - \ln 15] = 6/7$

$\ln((7p+1)/15) = 2 \rightarrow (7p+1)/15 = e^2$

$p = (15e^2 - 1)/7$

7 Segments & Newton–Raphson · semicircle, ∠BOC = θ — 7 marks

(a) Show $p\sin\theta + q\theta - 2\pi = 0$

∠AOB = $\pi - \theta$. Segment area = $\tfrac{1}{2}r^2(\text{angle} - \sin \text{angle})$.

$R_1 = \tfrac{1}{2}r^2(\theta - \sin\theta)$, $R_2 = \tfrac{1}{2}r^2((\pi-\theta) - \sin\theta)$ [$\sin(\pi-\theta)=\sin\theta$]

$R_1 = 2R_2$: $\theta - \sin\theta = 2(\pi - \theta - \sin\theta)$

$\theta - \sin\theta = 2\pi - 2\theta - 2\sin\theta \rightarrow 3\theta + \sin\theta - 2\pi = 0$

$p = 1, q = 3$

(b) Newton–Raphson, $\theta_0 = 1.3$

$f(\theta) = \sin\theta + 3\theta - 2\pi$, $f'(\theta) = \cos\theta + 3$

$f(1.3) = -1.4196$, $f'(1.3) = 3.2675$

$\theta_1 = 1.3 - (-1.4196 / 3.2675)$

$\theta_1 = 1.734$ (3 d.p.)

8 Integration from first principles · $x e^{2x}$ — 5 marks

(a) Express the limit as an integral

$\int_0^3 x e^{2x}\,dx$

(b) Show it equals $Ae^6 + B$

By parts ($u = x$, $dv = e^{2x}dx$): $\tfrac{1}{2}x e^{2x} - \tfrac{1}{4} e^{2x}$

At 3: $(3/2 - 1/4)e^6 = (5/4)e^6$. At 0: $-1/4$.

$(5/4)e^6 + 1/4 \rightarrow A = 5/4, B = 1/4$

9 Trig identity & equation

8 marks

(a) Show $(\cos\theta + \sin\theta)(\operatorname{cosec}\theta - \sec\theta) \equiv k \cot 2\theta$

$= (\cos\theta + \sin\theta)\cdot(\cos\theta - \sin\theta)/(\sin\theta \cos\theta) = (\cos^2\theta - \sin^2\theta)/(\sin\theta \cos\theta)$

$= \cos 2\theta \,/\, (\tfrac{1}{2} \sin 2\theta) = 2 \cot 2\theta$

$k = 2$

(b) Solve $5(\cos x+\sin x)(\operatorname{cosec} x-\sec x) = 4\operatorname{cosec}^2 2x$, $-90° < x < 90°$

$10 \cot 2x = 4(1 + \cot^2 2x) \rightarrow 2\cot^2 2x - 5\cot 2x + 2 = 0$

$(2\cot 2x - 1)(\cot 2x - 2) = 0 \rightarrow \cot 2x = \tfrac{1}{2}$ or 2

$\tan 2x = 2 \rightarrow 2x = 63.43°, -116.57°$

$\tan 2x = \tfrac{1}{2} \rightarrow 2x = 26.57°, -153.43°$

$x = -76.7°, -58.3°, 13.3°, 31.7°$ (1 d.p.)

10 Normal & area · $y = 7x / \sqrt{3x^2 + 4}$, P at $x = 2$

11 marks

(a) Equation of normal l

Differentiate: $dy/dx = 28(3x^2 + 4)^{-3/2}$

At $x = 2$: $16^{3/2} = 64 \rightarrow dy/dx = 28/64 = 7/16$.   P = (2, 7/2).

Normal gradient = $-16/7$: $y - 7/2 = -(16/7)(x - 2)$

$32x + 14y - 113 = 0$

(b) y-coordinate of Q (l meets y-axis)

$x = 0 \rightarrow 14y = 113$ **$y = 113/14$**

(c) Exact area of R

Area under line: $\int_0^2 (113 - 32x)/14 \, dx = 162/14 = 81/7$

Area under curve: $\int_0^2 7x/\sqrt{3x^2+4} \, dx = (7/3)[\sqrt{16} - \sqrt{4}] = 14/3$

$R = 81/7 - 14/3 = (243 - 98)/21$

$R = 145/21$

11 R cos(x+α) & electricity model

11 marks

(a) $4\cos x - 13\sin x = R\cos(x + \alpha)$

$R = \sqrt{(4^2 + 13^2)} = \sqrt{185}$, $\tan\alpha = 13/4$

$R = \sqrt{185}$, $\alpha = 1.273$ rad (3 d.p.)

(b) Minimum demand & time

$D = 30 + \sqrt{185}\cos(\pi t/12 + 0.2 + \alpha)$; min when cos = −1.

$D_{min} = 30 - \sqrt{185}$ **≈ 16.4 GW**

Occurs when $\pi t/12 + 1.479 = \pi \to t \approx 6.35$ **≈ 06:21**

(c) Rate k at 3 pm (t = 15)

$dD/dt = (\pi/12)[-4\sin(\pi t/12+0.2) - 13\cos(\pi t/12+0.2)]$

At t = 15: **k ≈ 2.76 GW/hr (3 s.f.)**

(d) Why not the same model every day?

Demand changes with the **seasons** (temperature, daylight hours), so fixed constants can't model every day of the year.

12 Modulus graph & intersections · $y = |px - q|$, $y = k/x$

7 marks

(a) Sketch of y = |px − q|

V-shape, vertex on x-axis at $(q/p, 0)$, crossing the y-axis at $(0, q)$.

(b) Range of k for 3 distinct intersections

y = k/x (k > 0) only meets the V for x > 0. Right branch always gives 1 point.

Left branch: $q - px = k/x \to px^2 - qx + k = 0$ needs 2 distinct positive roots → discriminant $q^2 - 4pk > 0$.

$0 < k < q^2/(4p)$

13 Geometric sequence · 3rd, 4th, 5th terms = sinθ, √2 cosθ, √3 cotθ

7 marks

(a) Common ratio

Equal ratios: $\sqrt{2}\cos\theta / \sin\theta = \sqrt{3}\cot\theta / (\sqrt{2}\cos\theta)$ → $\cos\theta = \sqrt{3}/2 \to \theta = \pi/6$

$r = \sqrt{2}\cos\theta / \sin\theta = \sqrt{2}\cdot(\sqrt{3}/2)/(1/2)$

$r = \sqrt{6}$

(b) Sum of first 3 terms

3rd term = sin(π/6) = ½, so term1 = ½/r² = 1/12, term2 = ½/r = √6/12.

$1/12 + \sqrt{6}/12 + 6/12$

$(7 + \sqrt{6})/12$

14 Differential equation · $dx/dt = x(A - t)$

11 marks

(a) Particular solution

Separate: $\int dx/x = \int (A - t)dt \rightarrow \ln x = At - t^2/2 + C$

x = 0.3 at t = 0 → C = ln 0.3. $x = 0.3\, e^{At - t^2/2}$

x = 21 at t = 4 → $e^{4A-8} = 70 \rightarrow A = 2 + \frac{1}{4}\ln 70$ (≈ 3.062)

$x = 0.3\, e^{At - t^2/2}$, $A = 2 + \frac{1}{4} \ln 70$

(b) Maximum concentration

Max when $dx/dt = 0 \rightarrow t = A$. $x_{max} = 0.3\, e^{A^2/2}$

≈ 32.6 mg L^{-1} (3 s.f.)

(c) Value of T when x = 0.1

$e^{AT - T^2/2} = 1/3 \rightarrow T^2 - 2AT - 2\ln 3 = 0 \rightarrow T = A + \sqrt{A^2 + 2\ln 3}$

T ≈ 6.46 (3 s.f.)

15 Proof by contradiction · $a^2 + b^2 = c^2$

4 marks

Complete the proof that a and b cannot both be odd

Assume $a = 2m+1$, $b = 2n+1$. Then

$a^2 + b^2 = 4(m^2 + m + n^2 + n) + 2 = c^2$

So c^2 is even → c is even → $c = 2p \rightarrow c^2 = 4p^2$ (divisible by 4).

But $c^2 = 4(\ldots) + 2 \equiv 2 \pmod 4$, **not** divisible by 4.

Contradiction → a and b cannot both be odd. ∎

Reading your mark scheme

- **"Show that" questions** give marks for the **method line**, not just the answer — full working on Q7(a), Q8(b), Q9(a), Q15 is where the marks live.
- **Accuracy:** watch rounding instructions — 3 d.p. vs 3 s.f. vs exact form. Losing the final A-mark to rounding is the most common drop.
- **"Solutions relying on calculator technology are not acceptable"** means you must show algebra/calculus — a typed answer scores zero even if correct.